capable of shelling out money. To take forward this initiative, SII signed a licence with British-Swedish drug maker AstraZeneca and Oxford University to produce one billion doses, out of which 400 million were to be delivered before the end of 2020.[1] These doses would be reserved exclusively for low- and middle-income countries and half a million of them would go to India each year. Oxford University and AstraZeneca had already begun Phase I and II human trials in the UK way back in April 2020 and had progressed to Phase III trials in May. Just last month, in June, they had begun Phase III trials in Brazil and Phase I/II trials in South Africa as well.

As per best scientific practices, bridge trials are crucial in understanding how a potential vaccine candidate works on the local population, considering varying issues of ethnicity and genetic variations. They are especially important early on to observe adverse effects, before the vaccine has been safely administered on a large scale across a wide range of ethnicities. Considering this, if SII had to apply for emergency use authorization in India, it would have to conduct these trials and furnish the outcomes, despite its partner companies having conducted trials elsewhere in the world. In view of this, the ICMR extended ₹11 crore worth of financial assistance to SII to conduct bridge trials in India in addition to the help provided in recruitment of 1,600 candidates for the trials.[2]

Bharat Biotech's vaccine candidate Covaxin, which was being jointly developed with the ICMR, had also published positive results from its pre-clinical trials and had begun Phase I trials in the country. Phases I and II of the trials generally observe for mild and adverse effects in a limited number of people against a placebo vaccine. ICMR provided ₹35 crore to Bharat Biotech for the trials on 25,800 participants.[3] The Zydus Cadila vaccine was also provided ₹10 crore to initiate and facilitate human trials.

Vaccine development process is long and complex around the world. Typically, it takes about a decade to research, develop and test a new vaccine. However, the fact that the scientific community came forth to produce not just one, but several vaccine candidates, while adhering to best practices, is absolutely commendable. The three vaccine candidates in India which looked promising at the moment were based on different scientific theories. Bharat Biotech's Covaxin is based on an inactivated virus platform. An inactivated pathogen does not multiply and cannot cause disease but is capable of producing an immune response in the body, thus